DESCRIPTION OF COURSEWORK

Course Code	AIT303
Course Name	Advanced Issues of Artificial Intelligence (Natural Language Processing)
Lecturer	Chua Chong Chai
Academic Session	2026/04
Assessment Title	Assignment 1

A. Introduction/ Situation/ Background Information

The assignment evaluates the knowledge and skills of the students to complete the full cycle of aspects-based sentiment analysis using sentiment analysis and topic modeling techniques. The student will apply latest techniques in natural language processing such as: text preprocessing, feature extraction, word embedding, machine learning (supervised, unsupervised and semi-supervised), and deep neural network to complete the assignment.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 1 Illustrate the advanced and latest issues of artificial intelligence in Natural Language Processing and their latest solutions. (C4, PLO2)

CLO 2 Analyse the advanced Natural Language Processing for industrial applications. (A4, PLO11)

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating

published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This assignment is individual basis.
2. The deadline of this assignment submission is on **5 June 2025 (Friday), 5pm**.
3. Complete all the programming tasks using Python and Jupyter Notebook.
4. For submission
    a. Submit the assignment report (in pdf format) online through Moodle;
    b. Upload all the Jupyter Notebooks (ipynb files), generated excel files and trained models (zip the files) to OneDrive/Google/Box shared folder. Highlight the link to this shared folder in the assignment report.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Assignment (Code and Report)	20
	TOTAL	**20**

F. Task(s)

Complete the following tasks:

1. Sentiment Analyzer
    a. Train prediction models to **predict the sentiment label (positive or negative)** for text using following algorithms:
        i. Support Vector Machine (SVM)
        ii. Bidirectional Gated Recurrent Unit (BiGRU)
    b. Use the **IMDB Movie Dataset** from Kaggle to train the models: https://www.kaggle.com/datasets/lakshmi25npathi/imdb-dataset-of-50k-movie-reviews
    c. Design and code the workflow for data preparation, text preprocessing, features extraction (or word embedding), model training, and model evaluation using Python.
    d. For the **SVM**, use **CountVectorizer** and **TfidfVectorizer** from **Scikit-Learn** for feature extraction to train two different models.
    e. For the **BiGRU**, use the **CBOW** and **Skip Gram** from **Gensim** for word embedding to train two different models.
    f. Use **cross-validation** techniques to validate and improve the models training for both of the **SVM** and **BiGRU**.
    g. Evaluate all the trained models in 1.d, 1.e and 1.f.
    h. Save all the trained models.

2. Aspect-based Sentiment Analysis
    a. Aspect Extraction
        i. Scrap online reviews of at least **30 different products** within the same category (will be assigned during the lecture class) from **Best Buy** (https://www.bestbuy.com). (*Make sure that each product has at least 100 reviews.)
        ii. Use the scrapped reviews to perform the following unsupervised and semi-supervised aspect extraction models training.
        iii. **Unsupervised aspect extraction**: Train aspect extraction models using LDA and BERTopic.

iv. **Semi-supervised aspect extraction**:
   - Analyze the **topics and key terms** generated from the LDA and BERTopic models.
   - Restructure the **key terms** into at least **6 aspects (or topics)** according to the relatedness of these key terms. (*These aspects could be referring to the main considerations when the user purchased the products, such as the design, quality, comfortability, and etc. according to the nature of the product.)
   - Train aspect extraction model using **CorEx** algorithm based on the restructured aspects and key terms.

v. Save all the trained aspect extraction models.

b. Process and **label all the downloaded product reviews** using the trained **CorEx model and sentiment analysis model (**use the **best sentiment analysis model** from task 1).

c. Based on the labeled results, **rank and list the top 5 products** with highest total positive sentiment value according to each of the 6 aspects. (*You will produce 6 lists of top products.)

d. Choose **a best product** according to **your personal preferences** of the aspects and explain why.

3. Report

a. Organize the report into proper sections to discuss the completed tasks.

b. Sentiment Analysis

   i. Discuss the workflow to train the sentiment analysis models.

   ii. Compare and discuss the accuracy of the **FOUR (4)** different models (show the details evaluation and confusion matrix).

   iii. Provide justifications for the model with the best accuracy.

c. Aspect-based Sentiment Analysis

   i. Observe and discuss the outputs generated from the **unsupervised** aspect extraction models.

ii. Discuss the aspects after the restructuring based on the results in the unsupervised aspect extraction.
iii. Observe and discuss the outputs generated from the **semi-supervised** aspect extraction model.
iv. Compare the outputs between the unsupervised and semi-supervised aspect extraction models.
v. Discuss the top 5 products according to each of the 6 aspects.
vi. Discuss the best product that you have chosen.
vii. For every aspect, create bar chart to illustrate the ranking of the top 5 products.

4. Code
   a. The code in Jupyter notebook should be documented (with comments) appropriately.

APPENDIX 1

MARKING RUBRICS

Component Title	**Assignment (Code and Report)**				**Percentage (%)**	**20**	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5-6)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Sentiment Analysis	**Student completed the full workflow and obtained excellent prediction results. No mistakes or errors in the coding and fulfil 100% of the assignment requirements.**	**Student completed the full workflow and obtained good prediction results. With minimum mistakes or errors in the coding and fulfil 90% of the assignment requirements.**	**Student fail to complete the machine learning workflow to train the prediction model. More than 70% but less than 90% of the processes in the workflow are performed.**	**Student fail to complete the machine learning workflow to train the prediction model. More than 30% but less than 70% of the processes in the workflow are performed.**	**Student fail to complete the machine learning workflow to train the prediction model. Less than 30% of the processes in the workflow are performed.**	**6**	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (8-9)	**Good (6-7)**	**Average (4-5)**	**Need Improvement (2-3)**	**Poor (1)**		
Aspect-based Sentiment Analysis	**Student completed the aspect-based sentiment analysis and obtained excellent results. No mistakes or errors in the coding and fulfil 100% of the assignment requirements.**	**Student completed the aspect-based sentiment analysis and obtained good results. With minimum mistakes or errors in the coding and fulfil 90% of the assignment requirements.**	**Student fail to complete the aspect-based sentiment analysis. More than 70% but less than 90% of the processes in the workflow are performed.**	**Student fail to complete the aspect-based sentiment analysis. More than 30% but less than 70% of the processes in the workflow are performed.**	**Student fail to complete the aspect-based sentiment analysis. Less than 30% of the processes in the workflow are performed.**	**9**	
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
The quality of Code & Report	**With complete documentation of the code. The report provides a concise summary of**	**With complete documentation of the code. The report provides a concise summary of**	**With simple documentation of the code. The report provides a summary of what is**	**With partial documentation of the code. The report provides a summary of what is**	**No proper documentation of the code. The report provides an unclear summary of**	**5**	

	what is accomplished with very detailed explanations and discussion. The report is properly formatted, easy to read and the content is accurate.	**what is accomplished with detailed explanations and discussion. The report is properly formatted and the content is accurate.**	**accomplished with explanations and discussion. The report has minor formatting problems. The content is accurate.**	**accomplished with partial explanation and discussion. Some part of the content is inaccurate. The report has some formatting problems.**	**what is accomplished, without explanation and discussion. The report is poorly formatted.**		
					TOTAL	**20**	

Note to students: Please include the marking rubric when submitting your coursework.